

UserSwarm — Pitch Script (Sponsor-tuned: Orkes ▶ Arize ▶ Terac)

Engineered against the **AI Hackathon 2026** rubric (Application · Functionality/Quality · Creativity · Technical Complexity · **Ethical Considerations** · **Brainstorming & Process**) and weighted toward the sponsor tracks in priority order: **Orkes first, then Arize, then Terac**. Format: science-fair table, ~3 min pitch + 2 min Q&A, judged $\geq 2\times$.

Spine of the story: *Orkes Conductor is what makes a swarm of AI users actually work* — it's the durable runtime that coordinates persona-generation, a parallel fan-out of browser testers, aggregation, and evals as one observable, fault-tolerant workflow. Arize proves the output is trustworthy; Terac calibrates it against humans.

0. One-liner (memorize this)

"UserSwarm releases a swarm of distinct AI users on your product and watches where they hit friction in a real browser — all coordinated as one durable Orkes workflow, scored live in Arize, and calibrated against human labels from Terac so the feedback is provably trustworthy."

Orkes-forward variant (lead with this at the Orkes table):

"We turned 'a swarm of AI users testing your product' from a script into a real orchestrated system — Orkes Conductor runs the whole multi-agent workflow durably and observably, so it survives failures, fans out testers in parallel, and you can watch every node execute."

1. The 3-minute pitch (timed)

Have the app open at `http://localhost:3000` with a finished Nimbus run ready, **Overview tab showing the Workflow Status timeline.**

[0:00–0:30] Hook + problem

"Every team ships UX they think is fine. Real user testing catches the friction — but it's slow and expensive. AI can give instant feedback, but two problems: why would you trust it, and how do you run a whole *swarm* of AI users reliably instead of a flaky script? UserSwarm solves both — and the reliability half is pure Orkes."

[0:30–1:00] What it does

"You give it a URL, what the product is, and who it's for. It generates a **swarm of distinct personas** — different patience, tech-savviness, disposition — and each one drives a **real headless browser** like an actual user. No test scripts."

[1:00–1:50] Demo — LEAD WITH ORKES (point at the Workflow Status timeline)

"This is the workflow, and it's a real **Orkes Conductor** orchestration — not a for-loop. See these engine badges? **Persona generation and aggregation run as durable Orkes agents**; the browser testers fan out in **parallel**, each in its own isolated session. Orkes coordinates the whole DAG — persona-gen → parallel testers → aggregate → evals — as one observable workflow. If a node fails, it surfaces as a real workflow error instead of silently faking a result. That's what makes 'a swarm' actually dependable."

Then the results (Results tab):

"Each AI user took ~30 real browser actions — replayable screenshots. They *disagree* like real users: this impatient one abandoned, this careful one pushed through, each with a voice-of-customer quote."

[1:50–2:30] Trust layer — Arize then Terac

"Top-right, these chips are live, not mocked — Claude, **Orkes**, **Arize**. And here's the trust part: **Arize traces every run and every LLM call**, and scores each report for **hallucination risk** and **evidence grounding** — eval-driven, not vibes. Then **Terac** human labels calibrate whether the AI agrees with real people — 87% agreement here. So you get a *trust score* on the feedback, not just more AI opinions."

[2:30–3:00] Payoff + close

"And it closes the loop: feed the human labels and failed evals back, improve the agent, rerun, and Arize shows a measurable before/after. Fast like AI, reliable because of Orkes, trustworthy because of Arize and Terac — and honest that it complements real research, not replaces it. That's UserSwarm."

2. Sponsor track strategy (priority: Orkes ▶ Arize ▶ Terac)

ORKES — make this the centerpiece

What they're judging: meaningful use of Conductor for **durable, observable, multi-agent orchestration** — not a thin API call.

Land these lines: - "The entire swarm is an **Orkes Conductor workflow**: persona-gen → **parallel** UX-tester fan-out → aggregation → evals, as one DAG." - "Reasoning agents run on the **Orkes durable runtime** (`AGENTSPAN_SERVER_URL`), so execution is **durable and observable** — every node's status streams into our timeline live." - "We made a **deliberate orchestration boundary**: tool-less reasoning agents run on Orkes workers; the live-browser testers run in-process because a Playwright session can't be shared across

workers. We can defend that trade-off." - **"No silent fallback** — if Orkes can't complete a node, the run raises a real error. We chose correctness/observability over hiding failures." *(This is a strong Orkes-philosophy point.)*

Show: the **Workflow Status timeline** with the **Orkes engine badges**, and call out the parallel tester fan-out.

Orkes Q&A: - "How are you using Conductor specifically?" → "Each reasoning agent is an Orkes agent/worker; the run is a workflow with a parallel fan-out stage and a join before aggregation. Status is observable per node." - "What happens on failure?" → "It surfaces as a workflow error, not a fake success — durability and honesty over silent degradation." - "Why Orkes and not just async?" → "Durability and observability. A swarm that re-runs on every deploy needs retrievable, inspectable execution — that's orchestration, not a loop."

ARIZE — the trust/observability proof

What they're judging: real **tracing + evals** driving the product, not a logging afterthought.

Land these lines: - "Arize traces every run and auto-instruments every Anthropic call via OpenInference — full LLM observability out of the box." - "We run **LLM-judge evals** (actionability, hallucination risk, evidence grounding) plus a **human-agreement eval**, logged to Arize as scored spans." - "Evals aren't decoration — they **gate the improve-and-rerun loop** and prove measurable before/after gains."

Show: the **Arize evals** grid (scores + explanations) and the trust chips.

Arize Q&A: "What do you trace?" → "Every workflow span and every LLM request, with eval scores as span attributes, exported over OTLP to Arize AX."

TERAC — human-in-the-loop calibration

What they're judging: real **human feedback** improving/validating the AI.

Land these lines: - "Terac collects **human labels** on the AI reports — useful-to-builder, specific-vs-vague, hallucinated, would-a-real-user-agree." - "Those labels power our **human-agreement eval** — the number that tells you whether to trust the swarm — and feed the **improve** step."

Honesty note: in the demo, Terac runs on **synthetic labels** (no live key), so present it as "the human-calibration layer, wired and computing agreement — swap in a Terac key and it's live." Don't claim it's live when the chip says mock.

3. Rubric map — one line per official category

Category	What to say / show
Application	"Any team with a web product — pre-launch UX QA in minutes, not a week of recruiting."
Functionality/Quality	Live run end-to-end, no crashes; live/mock chips prove real integrations; clean results UI.
Creativity	"Everyone builds AI that <i>generates</i> . We built an orchestrated, human-calibrated swarm — the novelty is reliability + trust, not the agents."
Technical Complexity	"Multi-agent DAG on the Orkes durable runtime , real Playwright automation with safety guardrails, LLM-judge + human-agreement evals , live OpenTelemetry tracing into Arize."
Ethical Considerations	§4 — privacy + honesty + auditability.
Brainstorming & Process	§5 — beats the "AI wrapper" critique.

4. Ethical Considerations (NEW category)

- **Privacy by construction.** Tests *your own* product with *synthetic* personas — no scraping real users, no PII.
 - **Honesty about limits.** The product says it: "*not a replacement for real user research.*"
 - **Trust, not blind automation.** Hallucination-risk + evidence-grounding evals (Arize) flag ungrounded claims; Terac human calibration checks the AI against people before you act.
 - **Safety guardrails.** Agents honor **do-not-click rules** (no delete-account / submit-payment on live sites).
 - **Transparency/auditability.** Every action is an **Orkes workflow node + Arize span + screenshot** — fully auditable, not a black box.
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5. Brainstorming & Process (NEW category — strongest differentiator)

Judges said it: *"not 100% vibe-coded apps or AI wrappers — show thought process and iteration."*

- **The core insight.** Not "make AI testers" — instead: *AI UX feedback is cheap but untrustworthy and hard to run reliably — fix both.* Orkes answers reliability; Arize+Terac answer trust.
 - **A deliberate architecture decision.** The Orkes/in-process boundary (reasoning on Orkes workers, live browser in-process) was a reasoned trade-off, not an accident.
 - **Real iteration we can show.** We caught the system **lying to itself** — agents reporting "field is broken" when our tooling just couldn't find unlabeled fields, and parallel testers overwriting each other's screenshots. We **diagnosed it from the Arize traces**, hardened the browser tooling (fuzzy field-matching, destructive-click guards, surfacing typed values), and re-ran to verify the false friction disappeared. That loop *is* the iterative process.
 - **The product embodies iteration.** Improve-and-rerun = observe → label → improve → re-measure.
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6. General Q&A (the 2 minutes after)

"Isn't this just an AI wrapper?"

"The agents are the easy part. The contribution is the system around them: **Orkes** makes the swarm a durable, observable workflow; **Arize** evals + **Terac** human labels make the output provably trustworthy; and an improve-and-rerun loop closes it."

"Can you trust AI user feedback?"

"Not blindly — that's the point. We quantify it: 87% human agreement here, low-grounding reports flagged. You see the trust score, not just the opinion."

"What's real vs mocked?"

"Claude, Orkes, and Arize are live — the chips poll the backend and would show 'mock' otherwise. Terac uses synthetic labels in the demo; it's wired to go live with a key."

"What's next?"

"A `select_option` browser tool so agents complete every flow, a regression mode that re-runs the swarm on each deploy as an Orkes-scheduled workflow, and a CI gate that fails a PR when friction spikes."

7. Pre-judging checklist

- [] Backend (`:8000`), frontend (`:3000`), Nimbus test app (`:3100`) all running.
- [] A **finished** run open on the **Overview tab** (Workflow timeline visible) — don't demo cold.
- [] Header chips showing **Claude · Orkes · Arize** live.
- [] At the **Orkes** table: open with the Orkes-forward one-liner (§0) and point at the timeline first.
- [] Devpost submitted as draft **before 11 AM**; **Orkes, Arize, Terac** sponsor tracks all selected.
- [] Whole team at the table; judges come to you.